## Section A: Pure Mathematics

**1** **(i)** Prove that, for any positive integers $n$ and $r$,

$$\frac{1}{{}^{n+r}\mathrm{C}_{r+1}} = \frac{r+1}{r}\left(\frac{1}{{}^{n+r-1}\mathrm{C}_{r}} - \frac{1}{{}^{n+r}\mathrm{C}_{r}}\right).$$

Hence determine

$$\sum_{n=1}^{\infty} \frac{1}{{}^{n+r}\mathrm{C}_{r+1}},$$

and deduce that $\displaystyle\sum_{n=2}^{\infty} \frac{1}{{}^{n+2}\mathrm{C}_{3}} = \frac{1}{2}$.

**(ii)** Show that, for $n \geqslant 3$,

$$\frac{3!}{n^3} < \frac{1}{{}^{n+1}\mathrm{C}_{3}} \quad \text{and} \quad \frac{20}{{}^{n+1}\mathrm{C}_{3}} - \frac{1}{{}^{n+2}\mathrm{C}_{5}} < \frac{5!}{n^3}.$$

By summing these inequalities for $n \geqslant 3$, show that

$$\frac{115}{96} < \sum_{n=1}^{\infty} \frac{1}{n^3} < \frac{116}{96}.$$

**Note:** ${}^{n}\mathrm{C}_{r}$ is another notation for $\binom{n}{r}$.

**2** The transformation $R$ in the complex plane is a rotation (anticlockwise) by an angle $\theta$ about the point represented by the complex number $a$. The transformation $S$ in the complex plane is a rotation (anticlockwise) by an angle $\phi$ about the point represented by the complex number $b$.

**(i)** The point $P$ is represented by the complex number $z$. Show that the image of $P$ under $R$ is represented by

$$\mathrm{e}^{\mathrm{i}\theta} z + a(1 - \mathrm{e}^{\mathrm{i}\theta}).$$

**(ii)** Show that the transformation $SR$ (equivalent to $R$ followed by $S$) is a rotation about the point represented by $c$, where

$$c \sin \tfrac{1}{2}(\theta + \phi) = a\,\mathrm{e}^{\mathrm{i}\phi/2} \sin \tfrac{1}{2}\theta + b\,\mathrm{e}^{-\mathrm{i}\theta/2} \sin \tfrac{1}{2}\phi,$$

provided $\theta + \phi \neq 2n\pi$ for any integer $n$.

What is the transformation $SR$ if $\theta + \phi = 2\pi$?

**(iii)** Under what circumstances is $RS = SR$?

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**3** Let $\alpha$, $\beta$, $\gamma$ and $\delta$ be the roots of the quartic equation

$$x^4 + px^3 + qx^2 + rx + s = 0\,.$$

You are given that, for any such equation, $\alpha\beta + \gamma\delta\,,\ \ \alpha\gamma + \beta\delta$ and $\alpha\delta + \beta\gamma$ satisfy a cubic equation of the form

$$y^3 + Ay^2 + (pr - 4s)y + (4qs - p^2s - r^2) = 0\,.$$

Determine $A$.

Now consider the quartic equation given by $p = 0\,,\ q = 3\,,\ r = -6$ and $s = 10\,.$

**(i)** Find the value of $\alpha\beta + \gamma\delta$, given that it is the largest root of the corresponding cubic equation.

**(ii)** Hence, using the values of $q$ and $s$, find the value of $(\alpha+\beta)(\gamma+\delta)$ and the value of $\alpha\beta$ given that $\alpha\beta > \gamma\delta\,.$

**(iii)** Using these results, and the values of $p$ and $r$, solve the quartic equation.

**4** For any function f satisfying $\mathrm{f}(x) > 0$, we define the *geometric mean*, F, by

$$\mathrm{F}(y) \ = \ \mathrm{e}^{\frac{1}{y}\int_0^y \ln \mathrm{f}(x)\,\mathrm{d}x} \qquad\qquad (y > 0)\,.$$

**(i)** The function f satisfies $\mathrm{f}(x) > 0$ and $a$ is a positive number with $a \neq 1$. Prove that

$$\mathrm{F}(y) = \ a^{\frac{1}{y}\int_0^y \log_a \mathrm{f}(x)\,\mathrm{d}x}\,.$$

**(ii)** The functions f and g satisfy $\mathrm{f}(x) > 0$ and $\mathrm{g}(x) > 0$, and the function h is defined by $\mathrm{h}(x) = \mathrm{f}(x)\mathrm{g}(x)$. Their geometric means are F, G and H, respectively. Show that $\mathrm{H}(y) = \mathrm{F}(y)\mathrm{G}(y)\,.$

**(iii)** Prove that, for any positive number $b$, the geometric mean of $b^x$ is $\sqrt{b^y}\,.$

**(iv)** Prove that, if $\mathrm{f}(x) > 0$ and the geometric mean of $\mathrm{f}(x)$ is $\sqrt{\mathrm{f}(y)}$, then $\mathrm{f}(x) = b^x$ for some positive number $b$.

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**5** The point with cartesian coordinates $(x, y)$ lies on a curve with polar equation $r = \mathrm{f}(\theta)$. Find an expression for $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of $\mathrm{f}(\theta)$, $\mathrm{f}'(\theta)$ and $\tan\theta$.

Two curves, with polar equations $r = \mathrm{f}(\theta)$ and $r = \mathrm{g}(\theta)$, meet at right angles. Show that where they meet

$$\mathrm{f}'(\theta)\mathrm{g}'(\theta) + \mathrm{f}(\theta)\mathrm{g}(\theta) = 0\,.$$

The curve $C$ has polar equation $r = \mathrm{f}(\theta)$ and passes through the point given by $r = 4$, $\theta = -\frac{1}{2}\pi$. For each positive value of $a$, the curve with polar equation $r = a(1 + \sin\theta)$ meets $C$ at right angles. Find $\mathrm{f}(\theta)$.

Sketch on a single diagram the three curves with polar equations $r = 1+\sin\theta$, $r = 4(1+\sin\theta)$ and $r = \mathrm{f}(\theta)$.

**6** *In this question, you are not permitted to use any properties of trigonometric functions or inverse trigonometric functions.*

The function T is defined for $x > 0$ by

$$\mathrm{T}(x) = \int_0^x \frac{1}{1+u^2}\,\mathrm{d}u\,,$$

and $T_\infty = \displaystyle\int_0^\infty \frac{1}{1+u^2}\,\mathrm{d}u$ (which has a finite value).

**(i)** By making an appropriate substitution in the integral for $\mathrm{T}(x)$, show that

$$\mathrm{T}(x) = \mathrm{T}_\infty - \mathrm{T}(x^{-1})\,.$$

**(ii)** Let $v = \dfrac{u+a}{1-au}$, where $a$ is a constant. Verify that, for $u \neq a^{-1}$,

$$\frac{\mathrm{d}v}{\mathrm{d}u} = \frac{1+v^2}{1+u^2}\,.$$

Hence show that, for $a > 0$ and $x < \dfrac{1}{a}$,

$$\mathrm{T}(x) = \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a)\,.$$

Deduce that

$$\mathrm{T}(x^{-1}) = 2\mathrm{T}_\infty - \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a^{-1})$$

and hence that, for $b > 0$ and $y > \dfrac{1}{b}$,

$$\mathrm{T}(y) = 2\mathrm{T}_\infty - \mathrm{T}\left(\frac{y+b}{by-1}\right) - \mathrm{T}(b)\,.$$

**(iii)** Use the above results to show that $\mathrm{T}(\sqrt{3}) = \frac{2}{3}\mathrm{T}_\infty$ and $\mathrm{T}(\sqrt{2}-1) = \frac{1}{4}\mathrm{T}_\infty$.

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**7** Show that the point $T$ with coordinates

$$\left(\frac{a(1-t^2)}{1+t^2}\,,\ \frac{2bt}{1+t^2}\right) \qquad (*)$$

(where $a$ and $b$ are non-zero) lies on the ellipse

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\,.$$

**(i)** The line $L$ is the tangent to the ellipse at $T$. The point $(X,Y)$ lies on $L$, and $X^2\neq a^2$. Show that

$$(a+X)bt^2-2aYt+b(a-X)=0\,.$$

Deduce that if $a^2Y^2>(a^2-X^2)b^2$, then there are two distinct lines through $(X,Y)$ that are tangents to the ellipse. Interpret this result geometrically. Show, by means of a sketch, that the result holds also if $X^2=a^2$.

**(ii)** The distinct points $P$ and $Q$ are given by $(*)$, with $t=p$ and $t=q$, respectively. The tangents to the ellipse at $P$ and $Q$ meet at the point with coordinates $(X,Y)$, where $X^2\neq a^2$. Show that

$$(a+X)pq=a-X$$

and find an expression for $p+q$ in terms of $a$, $b$, $X$ and $Y$.

Given that the tangents meet the $y$-axis at points $(0,y_1)$ and $(0,y_2)$, where $y_1+y_2=2b$, show that

$$\frac{X^2}{a^2}+\frac{Y}{b}=1\,.$$

**8** Prove that, for any numbers $a_1$, $a_2$, …, and $b_1$, $b_2$, …, and for $n\geqslant 1$,

$$\sum_{m=1}^{n}a_m(b_{m+1}-b_m)=a_{n+1}b_{n+1}-a_1b_1-\sum_{m=1}^{n}b_{m+1}(a_{m+1}-a_m)\,.$$

**(i)** By setting $b_m=\sin mx$, show that

$$\sum_{m=1}^{n}\cos(m+\tfrac{1}{2})x=\tfrac{1}{2}\big(\sin(n+1)x-\sin x\big)\operatorname{cosec}\tfrac{1}{2}x\,.$$

**Note:** $\sin A-\sin B=2\cos\big(\frac{A+B}{2}\big)\sin\big(\frac{A-B}{2}\big)$.

**(ii)** Show that

$$\sum_{m=1}^{n}m\sin mx=\big(p\sin(n+1)x+q\sin nx\big)\operatorname{cosec}^2\tfrac{1}{2}x\,,$$

where $p$ and $q$ are to be determined in terms of $n$.

**Note:** $2\sin A\sin B=\cos(A-B)-\cos(A+B)$;

$2\cos A\sin B=\sin(A+B)-\sin(A-B)$.

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## Section B: Mechanics

**9** Two particles $A$ and $B$ of masses $m$ and $2m$, respectively, are connected by a light spring of natural length $a$ and modulus of elasticity $\lambda$. They are placed on a smooth horizontal table with $AB$ perpendicular to the edge of the table, and $A$ is held on the edge of the table. Initially the spring is at its natural length.

Particle $A$ is released. At a time $t$ later, particle $A$ has dropped a distance $y$ and particle $B$ has moved a distance $x$ from its initial position (where $x < a$). Show that $y + 2x = \frac{1}{2}gt^2$.

The value of $\lambda$ is such that particle $B$ reaches the edge of the table at a time $T$ given by $T = \sqrt{6a/g}$ . By considering the total energy of the system (without solving any differential equations), show that the speed of particle $B$ at this time is $\sqrt{2ag/3}$ .

**10** A uniform rod $PQ$ of mass $m$ and length $3a$ is freely hinged at $P$.

The rod is held horizontally and a particle of mass $m$ is placed on top of the rod at a distance $\ell$ from $P$, where $\ell < 2a$. The coefficient of friction between the rod and the particle is $\mu$.

The rod is then released. Show that, while the particle does not slip along the rod,

$$(3a^2 + \ell^2)\dot{\theta}^2 = g(3a + 2\ell)\sin\theta\,,$$

where $\theta$ is the angle through which the rod has turned, and the dot denotes the time derivative.

Hence, or otherwise, find an expression for $\ddot{\theta}$ and show that the normal reaction of the rod on the particle is non-zero when $\theta$ is acute.

Show further that, when the particle is on the point of slipping,

$$\tan\theta = \frac{\mu a(2a - \ell)}{2(\ell^2 + a\ell + a^2)}\,.$$

What happens at the moment the rod is released if, instead, $\ell > 2a$?

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**11** A railway truck, initially at rest, can move forwards without friction on a long straight horizontal track. On the truck, $n$ guns are mounted parallel to the track and facing backwards, where $n > 1$. Each of the guns is loaded with a single projectile of mass $m$. The mass of the truck and guns (but not including the projectiles) is $M$.

When a gun is fired, the projectile leaves its muzzle horizontally with a speed $v - V$ relative to the ground, where $V$ is the speed of the truck immediately before the gun is fired.

**(i)** All $n$ guns are fired simultaneously. Find the speed, $u$, with which the truck moves, and show that the kinetic energy, $K$, which is gained by the system (truck, guns and projectiles) is given by

$$K = \tfrac{1}{2}nmv^2\left(1 + \frac{nm}{M}\right).$$

**(ii)** Instead, the guns are fired one at a time. Let $u_r$ be the speed of the truck when $r$ guns have been fired, so that $u_0 = 0$. Show that, for $1 \leqslant r \leqslant n$,

$$u_r - u_{r-1} = \frac{mv}{M + (n-r)m} \qquad (*)$$

and hence that $u_n < u$.

**(iii)** Let $K_r$ be the total kinetic energy of the system when $r$ guns have been fired (one at a time), so that $K_0 = 0$. Using $(*)$, show that, for $1 \leqslant r \leqslant n$,

$$K_r - K_{r-1} = \tfrac{1}{2}mv^2 + \tfrac{1}{2}mv(u_r - u_{r-1})$$

and hence show that

$$K_n = \tfrac{1}{2}nmv^2 + \tfrac{1}{2}mvu_n.$$

Deduce that $K_n < K$.

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## Section C: Probability and Statistics

**12** The discrete random variables $X$ and $Y$ can each take the values $1, \ldots, n$ (where $n \geqslant 2$). Their joint probability distribution is given by

$$\mathrm{P}(X = x,\ Y = y) = k(x + y)\,,$$

where $k$ is a constant.

**(i)** Show that

$$\mathrm{P}(X = x) = \frac{n + 1 + 2x}{2n(n + 1)}\,.$$

Hence determine whether $X$ and $Y$ are independent.

**(ii)** Show that the covariance of $X$ and $Y$ is negative.

**13** The random variable $X$ has mean $\mu$ and variance $\sigma^2$, and the function V is defined, for $-\infty < x < \infty$, by

$$\mathrm{V}(x) = \mathrm{E}\big((X - x)^2\big).$$

Express $\mathrm{V}(x)$ in terms of $x$, $\mu$ and $\sigma$.

The random variable $Y$ is defined by $Y = \mathrm{V}(X)$. Show that

$$\mathrm{E}(Y) = 2\sigma^2. \qquad (*)$$

Now suppose that $X$ is uniformly distributed on the interval $0 \leqslant x \leqslant 1$. Find $\mathrm{V}(x)$. Find also the probability density function of $Y$ and use it to verify that $(*)$ holds in this case.

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